Type MS · Decomposition and abstraction

Answer 1 of 3 · Question 4 part b · 2 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part b in this mark scheme extract

4(a)	A1	Breaking problem into small parts		1

Question	mp	Answer	Additional guidance	Mark
4(b)		Award **two** marks for a linked explanation such as:		
	B1 B2	• Programmer can ignore unnecessary detail (1) because it doesn't matter how library generates the number (1) • Eli can focus on just his own code (1) because library hides the code that generates the random numbers (1) • It doesn't matter how the subprogram generates the random number (1) because Eli only needs the number itself (1)	First mark for meaning of abstraction (focus on necessary parts / hiding unnecessary detail) Second mark for linking use of **random** library Do **not** accept answers relating to modelling real-world scenarios as this is in the question	2

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Decomposition and abstraction

Answer 1 of 3 · Question 4 part b · 2 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
4(c)		Award **one** mark for each of:		
	C1	Booster variable initialised to 1 (1)	booster = 1	
	C2	At least one given constant used correctly (1)		
	C3	Loop used to make three throws (1)	Either type	
	C4	Use of random.randint( ) to assign roll a value between 1 and 6 (1)	roll = random.randint(1,SIDES) roll = random.randint(1,6) Any accurate method do **not** accept: random.randint(6) random.randint(SIDES)	

	C5	Check for both 3 or 6 for at least one roll (1)	any method e.g.
			`if (roll == 3 or roll == 6)`
			`if roll % 3 == 0`	

	C6	booster incremented (1)	Must be indented under condition in C5. Can award even if C5 not functioning correctly, but must be idea of only boosting on 3 and 6	
	C7	Correct final score calculated from 3 random rolls (1)	Even if not displayed Does not need to use given variable (finalScore) to get this mark	**8**

14 of 23

Type MS · Decomposition and abstraction

Answer 1 of 3 · Question 4 part b · 2 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

	C8	Correct output strings including each roll score and final score with appropriate strings (1)	All three throws and final score as shown in figure 6. `roll` and `finalScore` variables should match candidate solution Allow sensible alternate strings e.g. Roll for Rolled or Total for Score – Ignore : (colon)	

15 of 23

Type MS · Decomposition and abstraction

Answer 1 of 3 · Question 4 part b · 2 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# Q04c

# import libraries
import random

# program constants
SIDES = 6
THROWS = 3

# complete the lines to initialise the variables
roll = 0
subtotal = 0
booster = 1
finalScore = 0

# roll dice 3 times
for throw in range(THROWS):
    # display number rolled
    roll = random.randint(1,SIDES)
    print("Rolled:",roll)

    # update subtotal
    subtotal = subtotal + roll

    # check for roll of 3 or 6 and increment booster if needed
    if roll == 3 or roll == 6:
        booster = booster + 1

# calculate and output final score
finalScore = subtotal * booster
print("Final score:", finalScore)
```

16 of 23

Type MS · Decomposition and abstraction

Answer 1 of 3 · Question 4 part b · 2 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part b in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 1 of 16 · find part a in this mark scheme extract

1(a)	A1	**The only correct answer is D** ***A*** *is not correct because it is the definition of decomposition* ***B*** *is not correct because it is a task undertaken during evaluation* ***C*** *is not correct because it describes a method for expressing algorithms*		**1**

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 2 of 16 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 3 of 16 · find part a in this mark scheme extract

1b		Award **one** mark for each of:		
	B1	Initialisation of myName to "" (1)	• `String myName = "";` • `String myName = "";` • `myName = ""` Allow null Allow single space between quotes " "	
	B2	Add a line to create myAge and initialise it to 0 (1)	• `int myAge = 0;` • `int myAge = 0;` • `myAge = 0` Allow any method	
	B3	Complete the test by inserting < (1)	• `if (myAge < 30)` • `if (myAge < 30)` • `if (myAge < 30):` Allow <= Can award reverse test only if welcome message contained within an 'else' part	
	B4	Add a line to display "Welcome" (1)	• `Console.WriteLine ("Welcome");` • `System.out.println ("Welcome");` • `print ("Welcome")` ignore indentation Allow any method	**4**

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 4 of 16 · find part a in this mark scheme extract

C#

```
// Q01bFINISHED

// Complete the line to initialise myName to an empty string
String myName = "";

// Add a line to create an integer variable called myAge and
//     initialise myAge to 0
int myAge = 0;

Console.WriteLine ("Enter your name: ");
myName = Console.ReadLine();
Console.WriteLine ("Enter your age: ");
myAge = Convert.ToInt32 (Console.ReadLine());

// Complete the test to check if the age is less than 30
if (myAge < 30)
{
    // Add a line to display a message that says "Welcome"
    Console.WriteLine ("Welcome");
}

```

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 5 of 16 · find part a in this mark scheme extract

Java

```java
// Q01bFINISHED

import java.util.Scanner;
public class Q01bFINISHED
{
    public static void main (String[] args)
    {
        Scanner myKeyboard = new Scanner (System.in);

        // Complete the line to initialise myName to an empty string
        String myName = "";

        // Add a line to create an integer variable called myAge and
        //     initialise myAge to 0
        int myAge = 0;

        System.out.println ("Enter your name: ");
        myName = myKeyboard.nextLine();
        System.out.println ("Enter your age: ");
        myAge = Integer.parseInt (myKeyboard.nextLine());

        // Complete the test to check if the age is less than 30
        if (myAge < 30)
        {
            // Add a line to display a message that says "Welcome"
            System.out.println ("Welcome");
        }
    }
}
```

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 6 of 16 · find part a in this mark scheme extract

Python

```python
# Q01bFINISHED

# Complete the line to initialise myName to an empty string
myName = ""

# Add a line to create an integer variable called myAge and
#     initialise myAge to 0
myAge = 0

myName = input ("Enter your name: ")
myAge = int (input ("Enter your age: "))

# Complete the test to check if the age is less than 30
if (myAge < 30):

    # Add a line to display a message that says "Welcome"
    print ("Welcome")

```

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 7 of 16 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 8 of 16 · find part a in this mark scheme extract

1c(i)		Award **one** mark for any of the following:		
	C1	C#: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Java: • // Q01c (1) • Q01c (1) • // Loop for 50 times (1) • Loop for 50 times (1) Python: • # Q01c (1) • Q01c (1) • # Loop for 50 times (1) Loop for 50 times (1)	Ignore transcription errors	**1**

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 9 of 16 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 10 of 16 · find part a in this mark scheme extract

1c(ii)		Award **one** mark for any of the following:		
	C2	C#: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Java: • int count = 0; (1) • int total = 0; (1) • int subtotal = 0; (1) Python: • count = 0 (1) • total = 0 (1) • subtotal = 0 (1)	Ignore transcription errors	1
Question	**mp**	**Answer**	**Additional guidance**	**Mark**

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 11 of 16 · find part a in this mark scheme extract

1c(iii)		Award **one** mark for:		
	C3	• if (1)	Applies to all programming languages Ignore capitalisation	**1**

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 12 of 16 · find part a in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 13 of 16 · find part a in this mark scheme extract

1c(iv)		Award **one** mark for:		
	C4	C#: • && (1) Java: • && (1) Python: • and (1)	Ignore capitalisation	1

Question	mp	Answer	Additional Guidance	Mark

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 14 of 16 · find part a in this mark scheme extract

1(d)	D1	**The only correct answer is B** ***A*** *is not correct because the data can be stored in an array as it is all the same data type (string)* ***C*** *is not correct because the data can be stored in an array as it is all the same data type (integer)* ***D*** *is not correct because the data can be stored in an array as it is all the same data type (real)*		**1**

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 15 of 16 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Decomposition and abstraction

Answer 2 of 3 · Question 1 part a · 1 mark

2024 JUN · Paper 02 · extract 16 of 16 · find part a in this mark scheme extract

<table>
<tr>
<td>1(e)</td>
<td>E1
E2</td>
<td>One mark for each correct cell:
<table>
<tr><th>Item</th><th>Data type</th></tr>
<tr><td>45.82</td><td>All programming languages:
real/float
C#:
decimal/double
Java:
Double/double/BigDecimal
Python:
decimal</td></tr>
<tr><td>True</td><td>All programming languages:
Boolean/bool</td></tr>
</table></td>
<td>Ignore capitalisation
Do not accept string for True</td>
<td>2</td>
</tr>
</table>

Question	mp	Answer	Additional Guidance	Mark

Type MS · Decomposition and abstraction

Answer 3 of 3 · Question 4 part a · 1 mark

2025 JUN · Paper 02 · extract 1 of 5 · find part a in this mark scheme extract

4(a)	A1	Breaking problem into small parts		1

Question	mp	Answer	Additional guidance	Mark
4(b)		Award **two** marks for a linked explanation such as:		
	B1 B2	• Programmer can ignore unnecessary detail (1) because it doesn't matter how library generates the number (1) • Eli can focus on just his own code (1) because library hides the code that generates the random numbers (1) • It doesn't matter how the subprogram generates the random number (1) because Eli only needs the number itself (1)	First mark for meaning of abstraction (focus on necessary parts / hiding unnecessary detail) Second mark for linking use of **random** library Do **not** accept answers relating to modelling real-world scenarios as this is in the question	2

Type MS · Decomposition and abstraction

Answer 3 of 3 · Question 4 part a · 1 mark

2025 JUN · Paper 02 · extract 2 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
4(c)		Award **one** mark for each of:		
	C1	Booster variable initialised to 1 (1)	booster = 1	
	C2	At least one given constant used correctly (1)		
	C3	Loop used to make three throws (1)	Either type	
	C4	Use of random.randint( ) to assign roll a value between 1 and 6 (1)	roll = random.randint(1,SIDES) roll = random.randint(1,6) Any accurate method do **not** accept: random.randint(6) random.randint(SIDES)	

	C5	Check for both 3 or 6 for at least one roll (1)	any method e.g.
			`if (roll == 3 or roll == 6)`
			`if roll % 3 == 0`	

	C6	booster incremented (1)	Must be indented under condition in C5. Can award even if C5 not functioning correctly, but must be idea of only boosting on 3 and 6	
	C7	Correct final score calculated from 3 random rolls (1)	Even if not displayed Does not need to use given variable (finalScore) to get this mark	**8**

14 of 23

Type MS · Decomposition and abstraction

Answer 3 of 3 · Question 4 part a · 1 mark

2025 JUN · Paper 02 · extract 3 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

	C8	Correct output strings including each roll score and final score with appropriate strings (1)	All three throws and final score as shown in figure 6. `roll` and `finalScore` variables should match candidate solution Allow sensible alternate strings e.g. Roll for Rolled or Total for Score – Ignore : (colon)	

15 of 23

Type MS · Decomposition and abstraction

Answer 3 of 3 · Question 4 part a · 1 mark

2025 JUN · Paper 02 · extract 4 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

```
# Q04c

# import libraries
import random

# program constants
SIDES = 6
THROWS = 3

# complete the lines to initialise the variables
roll = 0
subtotal = 0
booster = 1
finalScore = 0

# roll dice 3 times
for throw in range(THROWS):
    # display number rolled
    roll = random.randint(1,SIDES)
    print("Rolled:",roll)

    # update subtotal
    subtotal = subtotal + roll

    # check for roll of 3 or 6 and increment booster if needed
    if roll == 3 or roll == 6:
        booster = booster + 1

# calculate and output final score
finalScore = subtotal * booster
print("Final score:", finalScore)
```

16 of 23

Type MS · Decomposition and abstraction

Answer 3 of 3 · Question 4 part a · 1 mark

2025 JUN · Paper 02 · extract 5 of 5 · find part a in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark